Determinants of child stunting in Latin America

"Evidence on low birth weight, small size at birth, household wealth and altitude of residence as determinants of child stunting in Latin America or Andean populations. Return a matrix with: authors, year, journal, DOI, country, design, sample size, determinant studied, and effect size. Limit to 2015-2026." *Generated by LeapSpace*

The strongest Latin American/Andean evidence in the retrieved set shows **household poverty/wealth** as a consistent determinant of child stunting, with **altitude** also implicated in Andean settings and **low birth weight/small size at birth** appearing as a risk marker in Brazil and Peru-related evidence, though not every study reports the same effect metric. [1] [2] [3] [4] [5] [6] [7] [8]

Matrix of relevant studies, 2015–2026

Authors	Year	Journal	DOI	Country	Design	Sample size	Determinant studied	Effect size
Not provided in retrieved evidence	2026	The Lancet Regional Health - Americas	10.1016/j.lana.2026.101501	Ecuador	Cross-sectional population survey	Not reported in excerpt	**Altitude of residence**; household poverty markers	Stunting/wasting prevalence was higher in rural and Andean areas and **increased with altitude**; poverty markers were associated with increased risk, but no adjusted numeric effect size was given in the excerpt. [1]
Not provided in retrieved evidence	2020	Advances in Science, Technology and Engineering Systems	10.25046/aj050318	Peru (central Andes, Tunanmarca)	Cross-sectional	52 mothers with children under 5	**Household wealth / economic level**	Weekly income <US$50 correlated with stunting (**r = 0.503**); weekly food expenditure <US$31 (**r = 0.648**); per-capita food expenditure <US$1.10/day (**r = 0.591**); no home ownership (**r = 0.413**). [2]
Not provided in retrieved evidence	2020	International Journal for Equity in Health	10.1186/s12939-020-01165-9	Latin America (13 countries)	Pooled analysis of nationally representative surveys	Not reported in excerpt	**Household wealth**	After adjustment for wealth and place of residence, indigenous children still had higher stunting prevalence than reference children (**PR = 1.34, 95% CI 1.28–1.39**). Wealth gradients were monotonic in all countries, but the excerpt does not give a single pooled wealth coefficient. [3] [9]
Not provided in retrieved evidence	2019	International journal for equity in health	10.1186/s12939-019-1016-0	Guatemala	Repeated national survey analysis	Five national health surveys (1995–2014)	**Household wealth**	The slope index for absolute wealth inequality in stunting ranged from **−52.9 to −60.4 percentage points**; rural children were consistently more stunted than urban children, and indigenous children in the poorest tertile had the highest prevalence. [4]
Not provided in retrieved evidence	2022	Child Indicators Research	10.1007/s12187-021-09880-7	Colombia	Decomposition analysis	2015 National Nutritional Situation Survey	**Household wealth**	Household wealth explained **54%** of the urban-rural stunting gap; the total gap was **7.2 percentage**

								points and the three main determinants together explained **92%** of the gap. [5]
Not provided in retrieved evidence	2020	BMC Public Health	10.1186/s12889-020-08869-1	Brazil (Alagoas)	Repeated cross-sectional household surveys	1231 (1992), 1381 (2005), 988 (2015)	**Low birth weight**	In 1992, low birth weight was associated with higher stunting prevalence (**28.3% vs 15.7%**); in 2015, stunting was again higher among low-birth-weight children (**8.6% vs 3.0%**). [7]
Not provided in retrieved evidence	2019	Jornal de Pediatria	10.1016/j.jped.2018.03.007	Argentina (Jujuy Province, Andean high-altitude setting)	Cross-sectional analysis of public facility birth records	48,656 newborns	**Altitude of residence**	Underweight, stunting, and wasting were all significantly higher at **>2000 m**; the excerpt reports prevalence of stunting at **3.39%** overall and states risks were higher at higher altitude, but does not provide an adjusted OR in the excerpt. [6]

Bottom line

Across the retrieved Latin American and Andean studies, **wealth-related disadvantage is the most consistently documented determinant of child stunting**, while **altitude shows a clear association in Andean/highland settings** and **low birth weight is an important risk marker where measured**, especially in Brazil; however, several abstracts report prevalence contrasts or correlations rather than fully adjusted effect estimates. [1] [2] [4] [5] [6] [7]

Confidence level: **High** — the retrieved evidence directly supports the wealth, altitude, and low-birth-weight patterns in Latin America/Andean populations, but the available excerpts often limit the precision of effect-size reporting.

References

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